

Allergen Control and Sanitation Procedure

Shared Production Lines (Line 3 / Line 4)

SOP Number	SOP-QA-014	Revision	3.2
Effective Date	March 1, 2026	Review Cycle	Annual
Document Owner	Quality Assurance Dept.	Approved By	Plant Quality Manager
Applicable Site(s)	Line 3 / Line 4 (Shared)	Classification	Internal - Controlled

1. Purpose

This procedure defines the required steps for cleaning, sanitizing, and verifying shared production lines between production runs involving different major food allergens. It is intended to reduce the risk of undeclared allergen cross-contact and to provide a documented, repeatable basis for line release decisions.

2. Scope

This SOP applies to all changeovers on Line 3 and Line 4 where a production run containing one or more of the nine major food allergens is followed by a run for a product with a different allergen profile, including allergen-free products. It covers dry clean-up, wet cleaning (CIP/COP), sanitization, and post-sanitation verification. It does not cover general equipment maintenance, which is addressed in SOP-ME-002.

3. Definitions

- Major Food Allergen: One of the nine allergens recognized under U.S. labeling law: milk, eggs, fish, crustacean shellfish, tree nuts, peanuts, wheat, soybeans, and sesame.
- Allergen Changeover: A cleaning and verification event that occurs specifically because the upcoming run has a different allergen profile than the preceding run.
- CIP (Clean-in-Place): Automated circulation of cleaning and sanitizing solutions through enclosed process equipment without disassembly.
- COP (Clean-out-of-Place): Manual removal, disassembly, and cleaning of parts outside the line, typically in a dedicated wash sink or parts washer.
- ATP (Adenosine Triphosphate) Swab: A rapid bioluminescence test used to verify general cleanliness of a surface, reported in Relative Light Units (RLU).
- Validated Cleaning: A cleaning method that has been documented, through allergen-specific testing, to reliably reduce allergen residue below the site action limit.

4. Responsibilities

Role	Responsibilities
Line Operator	Performs pre-cleaning teardown, removes gross soil, and completes the Line Clearance Checklist prior to handoff to Sanitation.
Sanitation Technician	Executes the wet-clean (CIP/COP) cycle, verifies chemical concentrations, and records contact times on the Sanitation Log.
QA Technician	Performs post-sanitation ATP swabbing and allergen surface testing; approves or rejects the line for allergen changeover release.
Shift Supervisor	Confirms changeover schedule, resolves scheduling conflicts between allergen

	runs, and co-signs the Line Release Form.
Plant Quality Manager	Owns this SOP, reviews deviation reports, and approves any temporary variance to stated parameters.

5. Materials and Equipment

- Alkaline CIP detergent, 1.5% to 2.0% w/v working solution
- Acid rinse additive, 0.5% w/v working solution (used on lines processing dairy soils)
- Peracetic-acid-based sanitizer, 150-200 ppm working concentration
- ATP luminometer and single-use swabs (calibrated per manufacturer schedule)
- Allergen-specific lateral flow or ELISA test kits (allergen-dependent; see Attachment B)
- Color-coded cleaning tools dedicated by allergen zone (per SOP-QA-009, Tool Color Coding)

6. Procedure

6.1 Pre-Cleaning Preparation

1. Line Operator confirms the production schedule and identifies the allergen status of the completed run and the upcoming run.
2. Remove all in-process product, packaging, and labels from the line; segregate for disposition per SOP-QA-011.
3. Complete the Line Clearance Checklist and route to the Shift Supervisor for initial sign-off.
4. Disconnect and stage disassembled parts for COP cleaning in the designated wash area.

6.2 Dry Clean-Up

5. Remove gross soil, product residue, and loose debris using dedicated dry tools (brushes, scrapers, vacuum).
6. Do not introduce water at this stage; wet debris is significantly harder to fully remove during CIP and increases allergen carryover risk.
7. Dispose of collected debris in allergen-labeled waste receptacles.

6.3 Wet Clean (CIP / COP)

8. Circulate a pre-rinse of potable water at 100–110°F for a minimum of 5 minutes to remove residual soil.
9. Circulate alkaline detergent solution at 1.5–2.0% w/v, 130–140°F, for a minimum contact time of 15 minutes.
10. For lines with heavy dairy or protein soiling, follow with an acid rinse additive at 0.5% w/v for 8 minutes.
11. Intermediate rinse with potable water until effluent runs clear and pH is within 6.5–8.5.
12. For COP items, submerge in the same detergent concentration for a minimum of 10 minutes with manual agitation, then rinse.

6.4 Allergen-Specific Changeover Steps

The following additional steps apply only when the upcoming run has a different major allergen profile than the completed run:

13. QA Technician reviews the changeover matrix in Attachment B to confirm whether the CIP/COP cycle above constitutes Validated Cleaning for the allergen pair involved.
14. If the changeover matrix indicates a non-validated pair, perform a second full detergent cycle (repeat step 6.3, items 2–4) before proceeding.
15. Visually inspect all product contact surfaces, gaskets, and dead-leg areas using a flashlight; document any residue found on the Visual Inspection Log.
16. Do not proceed to sanitization until visual inspection is passed and documented.

6.5 Sanitization

17. Circulate peracetic-acid-based sanitizer at 150–200 ppm for a minimum contact time of 60 seconds on all product contact surfaces.
18. Verify concentration using a test strip or titration prior to circulation; record the reading on the Sanitation Log.
19. Allow product contact surfaces to air dry; do not towel-dry with non-dedicated cloths.

6.6 Post-Sanitation Verification

20. QA Technician swabs a minimum of five predetermined product contact points using the ATP luminometer.
21. A reading below 250 RLU at each point is required to pass. Any point at or above 250 RLU requires re-cleaning of that surface and re-swabbing before the line may proceed.
22. For allergen changeovers identified as non-validated in step 6.4.1, QA Technician additionally performs allergen-specific surface testing per Attachment B. A result below the site action limit (10 ppm for total protein, allergen-specific kit dependent) is required to release the line.
23. QA Technician completes and signs the Line Release Form; Shift Supervisor co-signs before production may resume.

7. Corrective Actions

If any verification step in Section 6.6 fails, the following actions apply:

- The affected surface or the full line (QA Technician's determination, based on failure pattern) is re-cleaned per Section 6.3 before re-testing.
- Two consecutive failures at the same point trigger a documented deviation report and notification to the Plant Quality Manager before a third attempt.
- Any product run before a passing verification result is documented is placed on hold pending disposition review.
- Repeated failures (3 or more events in a rolling 30-day period) on the same line trigger a root-cause investigation per SOP-QA-020.

8. Documentation and Record Retention

The following records are generated by this procedure and retained by the QA Department for a minimum of 3 years from the date of creation: Line Clearance Checklist, Sanitation Log, Visual Inspection Log, ATP and allergen test results, Line Release Form, and any associated deviation reports.

9. Training Requirements

All personnel performing tasks under this SOP must complete initial allergen awareness and sanitation training and demonstrate competency through direct observation before working unsupervised. Refresher training is required annually and whenever this SOP is revised at a level that changes a stated parameter (concentration, time, or temperature).

10. Related Documents

- SOP-QA-009 – Tool Color Coding by Allergen Zone
- SOP-QA-011 – Product Disposition and Hold Procedure
- SOP-QA-020 – Root Cause Investigation Procedure
- SOP-ME-002 – Preventive Maintenance, Production Lines
- Attachment A – Line Clearance Checklist (form)
- Attachment B – Allergen Changeover Validation Matrix

11. Revision History

Rev	Date	Author	Description of Change
2.0	2023-06-14	QA Dept.	Added ATP swabbing threshold and dedicated allergen changeover section.
3.0	2025-01-09	QA Dept.	Updated sanitizer concentration range to align with supplier reformulation; added sesame as 9th major allergen.
3.1	2025-11-20	QA Dept.	Clarified corrective action timeline for failed swab results.
3.2	2026-03-01	QA Dept.	Extended record retention period from 2 to 3 years per updated corporate policy.

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